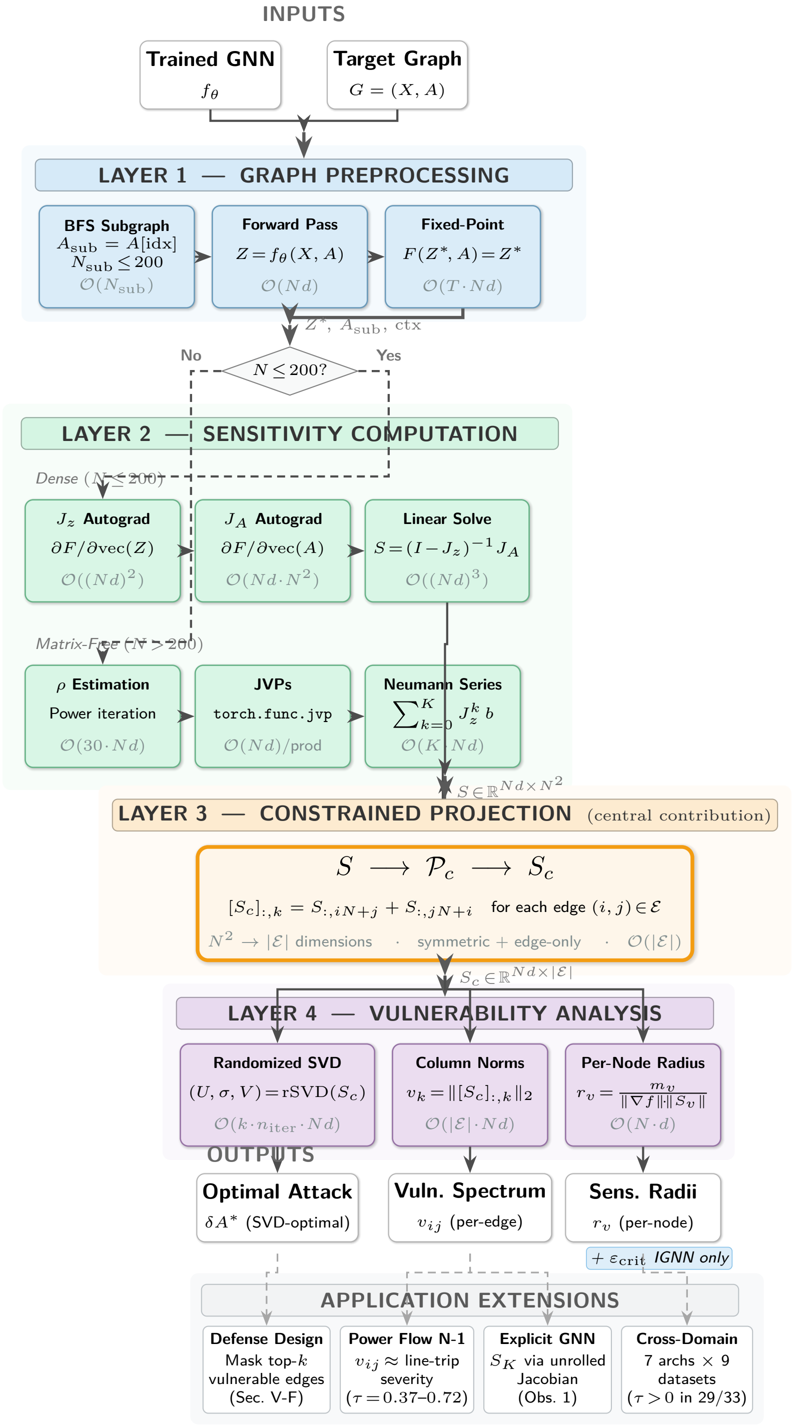
INPUTS
Trained GNN
f_θ
Target Graph
$G=(X,A)$
LAYER 1 — GRAPH PREPROCESSING
BFS Subgraph
$A_{\text{sub}}=A[\text{idx}]$
$N_{\text{sub}}\leq 200$
$\mathcal{O}(N_{\text{sub}})$
Forward Pass
$Z=f_\theta(X,A)$
$\mathcal{O}(Nd)$
Fixed-Point
$F(Z^*,A)=Z^*$
$\mathcal{O}(T\cdot Nd)$
$Z^*,\ A_{\text{sub}},\ \text{ctx}$
No
$N\leq 200?$
Yes
LAYER 2 — SENSITIVITY COMPUTATION
Dense ($N\leq 200$)
J_z Autograd
$\partial F/\partial \text{vec}(Z)$
$\mathcal{O}((Nd)^2)$
J_A Autograd
$\partial F/\partial \text{vec}(A)$
$\mathcal{O}(Nd\cdot N^2)$
Linear Solve
$S=(I-J_z)^{-1}J_A$
$\mathcal{O}((Nd)^3)$
Matrix-Free ($N>200$)
ρ Estimation
Power iteration
$\mathcal{O}(30\cdot Nd)$
JVPs
torch.func.jvp
$\mathcal{O}(Nd)$/prod
Neumann Series
$\sum_{k=0}^{K} J_z^k\, b$
$\mathcal{O}(K\cdot Nd)$
$S\in\mathbb{R}^{Nd\times N^2}$
LAYER 3 — CONSTRAINED PROJECTION (central contribution)
$S \longrightarrow \mathcal{P}_c \longrightarrow S_c$
$[S_c]_{:,k}=S_{:,iN+j}+S_{:,jN+i}$ for each edge $(i,j)\in\mathcal{E}$
$N^2\rightarrow|\mathcal{E}|$ dimensions · symmetric + edge-only · $\mathcal{O}(|\mathcal{E}|)$
$S_c\in\mathbb{R}^{Nd\times|\mathcal{E}|}$
LAYER 4 — VULNERABILITY ANALYSIS
Randomized SVD
$(U,\sigma,V)=\text{rSVD}(S_c)$
$\mathcal{O}(k\cdot n_{\text{iter}}\cdot Nd)$
Column Norms
$v_k=\|[S_c]_{:,k}\|_2$
$\mathcal{O}(|\mathcal{E}|\cdot Nd)$
Per-Node Radius
$r_v=\frac{m_v}{\|\nabla f\|\cdot\|S_v\|}$
$\mathcal{O}(N\cdot d)$
OUTPUTS
Optimal Attack
δA^* (SVD-optimal)
Vuln. Spectrum
v_{ij} (per-edge)
Sens. Radii
r_v (per-node)
$+\ \varepsilon_{\text{crit}}$ IGNN only
APPLICATION EXTENSIONS
Defense Design
Mask top-k vulnerable edges (Sec. V-F)
Power Flow N-1
$v_{ij}\approx$ line-trip severity ($\tau=0.37$–0.72)
Explicit GNN
S_K via unrolled Jacobian (Obs. 1)
Cross-Domain
7 archs × 9 datasets ($\tau>0$ in 29/33)